

$$= 1.66 \times 10^{-25} \text{ Nm (clockwise)}$$

6 (a)



(b)

$$C = C_0 e^{-\lambda t}$$

$$\ln C = -\lambda t + \ln C_0$$

time t / s	count rate C / s^{-1}	$\ln (C/\text{s}^{-1})$
2000	175	5.165
6000	28	3.332

$$-\lambda = \frac{3.332 - 5.165}{6000 - 2000}$$

$$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}} = 4.582 \times 10^{-4} \text{ s}^{-1}$$

$$t_{\frac{1}{2}} = 1510 \text{ s}$$

7(a)(i)

$$y = kMgl^r d^s b^{-1}$$